30538 Final Project Writeup

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Background and Research Question

This project examines the relationship between flood risk, as measured by the Special Flood Hazard Areas (SFHAs) designated by the Federal Emergency Management Agency (FEMA) and actual historical flood damage represented by National Flood Insurance (NFIP) claims, and risk aversion among home owners as measured by property values and purchase of NFIP policies. The analysis proceeds with the following research question in mind:

Does climate- and geography-related flood risk impact people's willingness to purchase flood insurance or the desirability of homes in areas with higher flood risk?

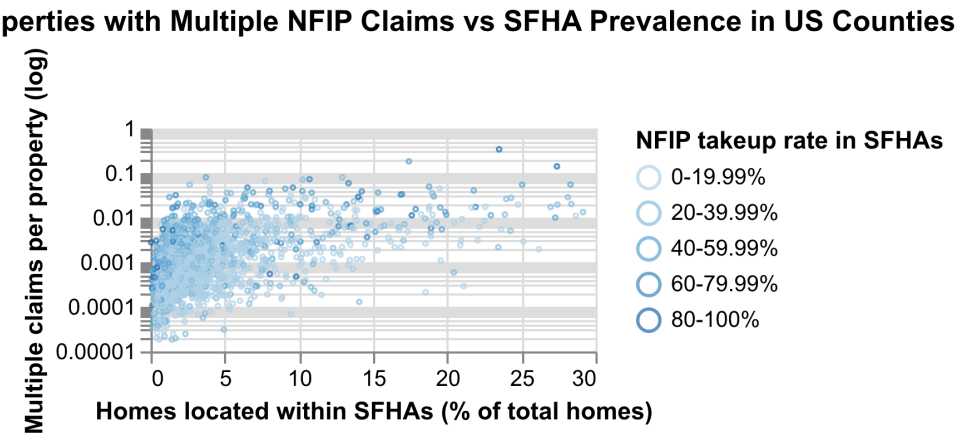
To investigate this, I used the following data sets: monthly county-level median housing value data from Zillow, a list of residential properties with multiple NFIP insurance claims, a list of residential NFIP takeup rates both in counties overall and in SFHAs, an official list of FEMA regions, and a shapefile of US states from the US Census Bureau. Property value trends over time were meant as a proxy for the relative desirability of living in a given location, while the NFIP multiple-loss property list was used as a proxy for susceptibility to flood damage. FEMA's SFHAs are also used to represent flood risk. NFIP takeup rates are meant to demonstrate whether people are aware of flood risk, as well as the strength of NFIP outreach in a given area. Because NFIP takeup rate and SFHA data was available only at the county level, other variables were also taken at the county level (rather than a more granular one). The data sets are matched on county (after adjusting for different naming conventions across organizations) and aggregated at the state and FEMA region level.

Challenges and Limitations

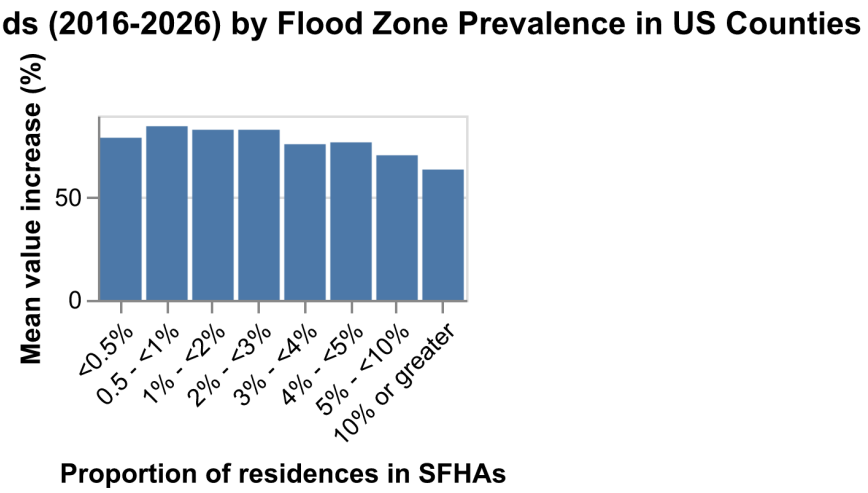
Before discussing the results, it is important to highlight a few limitations in the data. First, because of the lack of more granular NFIP takeup/SFHA data, we cannot see where intra-county variation is coming from or compare homes in SFHAs to homes outside of SFHAs. Also, data on claims and takeup rates includes only insurance purchased through the NFIP rather than all flood insurance. Finally, there are two states missing from parts of the analysis: Alaska because it was not included in the property value data, and Connecticut due to a 2023 change

in how Connecticut counties are structured, which made matching impossible due to the lack of 1:1 correspondence in regions that existed before and after the change.

Static Visualizations

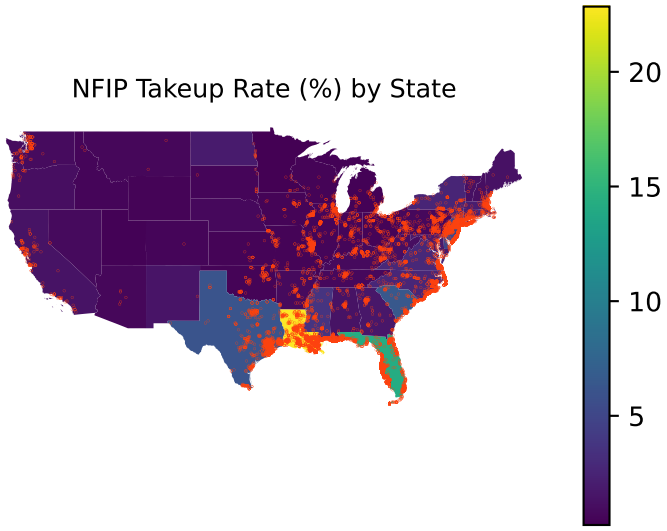


This chart displays a positive relationship between multiple-claim properties and the percentage of homes in a county in SFHAs, indicating that SFHAs do accurately predict flood damage. It also indicates that when NFIP takeup rates in SFHAs are lower, there tend to be slightly fewer claims, which could be because takeup rates are higher in more severe SFHAs or because there will naturally be fewer claims where there are fewer policies.



This second plot demonstrates a clear negative relationship between the percentage of homes in a county located in SFHAs and the trend in that county's residential property values over the past decade. The difference in mean property value trend that is associated with relatively small changes in the SFHA prevalence within a county is suggestive that climate-related flood risk (in other words, the increasing severity of the risk associated with a SFHAs) may be

priced in when homebuyers are determining where to purchase between otherwise comparable counties.



The final static visualization is a map displaying the percentage of residences in each state covered by a flood insurance plan under the NFIP. The orange dots represent properties with 5 or more NFIP insurance claims. Takeup rates are higher in states known for natural disasters like hurricanes (in particular Texas, Louisiana, and Florida), and the locations of properties with large numbers of claims are in many cases consistent with where we would expect high levels of hurricane-induced flood damage (such as the Gulf Coast) or flood damage related to rising sea levels (such as Cape Cod), which suggests people are thinking about climate when deciding whether to purchase flood insurance. However, there are many locations with high numbers of multiple-loss properties and low takeup rates, and takeup rates are fairly low everywhere, suggesting that there is room for more outreach.

Streamlit

The streamlit dashboard, available here, shows average property values and average growth in property values compared to the percentage of homes in a geographic area located in SFHAs. The user can choose to display the plots at the county, state, or FEMA region level, as well as choosing between the two outcome variables (property value and property value growth). The user can also choose whether to exclude outliers (functionally, this means counties with more than 20% of residences in SFHAs), which changes the range of the x-axis and makes variation at the lower end of the distribution easier to see. All of the plots show a general negative relationship between SFHA prevalence and both property value and growth in property values. This indicates both that longstanding flood risk (or something correlated with it) is likely priced into regional housing prices and that increasing flood risk as a result of climate change may be dynamically incorporated as property values continue to shift.